

# Customer Sentiment Classification Using a No-Code Deep Learning Neural Network

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**Abstract**—Deep learning has become an effective approach for text classification because it allows models to learn meaningful language patterns directly from examples instead of relying on manually written rules. This laboratory exercise explored the use of a no-code neural network to classify customer feedback into three categories: Positive Praise, Negative Complaint, and Urgent Support. A text classification model was developed using the Machine Learning for Kids platform, where labeled customer messages were collected, trained, and evaluated through multiple testing scenarios. The initial model was trained using ten examples per category and was later refined by expanding the dataset to twenty examples per category to observe how additional training data affected performance. The experiment demonstrated that increasing the amount and diversity of labeled examples generally improved the model's ability to recognize customer intent, although certain challenging cases, particularly sarcastic and context-dependent statements, continued to present classification difficulties. These observations highlight both the strengths and limitations of deep learning in natural language processing while emphasizing the importance of high-quality and representative datasets. Overall, the activity provided valuable insight into how neural networks learn from tagged examples and how iterative refinement contributes to building more reliable text classification models.

**Keywords**—Deep Learning, Neural Networks, Natural Language Processing, Text Classification, Sentiment Analysis, Machine Learning for Kids

## I. INTRODUCTION

Understanding customer feedback has become increasingly important as businesses receive large volumes of messages through websites, mobile applications, social media platforms, and online support channels. These messages often contain valuable information about customer satisfaction, complaints, or urgent concerns that require immediate attention. Manually reviewing every message can be both time-consuming and inconsistent, especially when the volume of feedback continues to grow. Because of this, organizations are increasingly turning to machine learning techniques that can automatically recognize patterns in text and classify customer concerns more efficiently.

Unlike traditional rule-based systems that depend on manually created keywords or numerous IF-ELSE statements, deep learning models learn directly from labeled examples. Rather than memorizing specific words, a neural network gradually discovers relationships between words,

phrases, and their corresponding categories during training. This ability allows the model to generalize beyond the exact examples it has seen, making it more adaptable to the different ways people naturally express their thoughts. At the same time, this exercise also illustrates that the quality of a model is closely tied to the quality of the data it learns from. A well-prepared and diverse dataset often contributes more to successful predictions than simply increasing computational complexity.

In this laboratory exercise, a no-code neural network was developed using the Machine Learning for Kids platform to classify customer feedback into three intent categories: **Positive Praise**, **Negative Complaint**, and **Urgent Support**. The activity involved creating a labeled dataset, training the model, evaluating its performance using unseen test cases, and refining the dataset to improve the classifier's performance. Beyond measuring accuracy, the exercise encouraged reflection on how neural networks learn from examples, why some predictions succeed while others fail, and how dataset quality influences the overall effectiveness of deep learning models. Through this process, the activity provided a practical introduction to one of the fundamental applications of natural language processing and demonstrated the iterative nature of developing reliable machine learning solutions.

## II. METHODOLOGY

This study followed a practical workflow to develop and evaluate a deep learning model capable of classifying customer feedback within an **E-commerce** environment. The Machine Learning for Kids platform was used to create a no-code text classification model, allowing the focus of the exercise to remain on understanding how a neural network learns from labeled data rather than on programming the underlying algorithms. The overall process consisted of dataset preparation, model training, performance evaluation, and model refinement.

### A. Dataset Preparation

An E-commerce business domain was selected because online retailers regularly receive large volumes of customer feedback through product reviews, customer support tickets, and service inquiries. These messages often express different intentions that businesses need to recognize

quickly in order to provide appropriate responses. Automating this process can improve customer service by helping prioritize urgent concerns while also identifying customer satisfaction and common complaints.

Three customer intent categories were defined for the dataset:

- **Positive\_Praise** – messages expressing satisfaction, appreciation, or positive experiences with products or services.
- **Negative\_Complaint** – messages describing dissatisfaction, delivery issues, defective products, or poor customer experiences.
- **Urgent\_Support** – messages requiring immediate assistance, such as payment problems, account access issues, or order-related emergencies.

During the initial stage, ten unique customer feedback statements were prepared for each category. Every example was manually labeled according to its corresponding intent before being added to the training dataset. Care was taken to create examples that reflected realistic situations commonly encountered in E-commerce platforms while avoiding duplicate or nearly identical statements. This diversity was important because the neural network relies on representative examples to learn meaningful language patterns rather than memorizing individual sentences.

### B. Model Training

After completing the dataset, the examples were submitted to the Machine Learning for Kids platform to train a neural network-based text classifier. During training, the system converted the labeled text into numerical representations and gradually learned the relationships between words, phrases, and their associated categories. Unlike traditional rule-based systems, no explicit keywords or IF-ELSE conditions were defined by the user. Instead, the model learned these relationships directly from the tagged examples provided during the training process.

The first training iteration was performed using **10 labeled examples per category**, producing an initial model that served as the baseline for evaluating classification performance. This initial model made it possible to observe how effectively the neural network generalized from a relatively small dataset before any refinements were introduced.

### C. Model Testing and Evaluation

The equations are an exception to the prescribed spTo evaluate the trained model, a separate collection of customer feedback messages that had not been included in the training dataset was prepared as the test set. These unseen messages represented different customer situations and writing styles to assess how well the neural network could classify new inputs instead of simply recognizing sentences it had already encountered.

The model's performance was measured by comparing its predicted category with the expected category for each test case. To evaluate the trained model, a separate collection of

customer feedback messages that had not been included in the training dataset was prepared as the test set. Each predicted category generated by the neural network was compared with its corresponding expected label to determine whether the classification was correct. These prediction results were later used to construct confusion matrices and compute the evaluation metrics presented in the Results and Discussion section.

### D. Model Refinement

After evaluating the initial model, additional training examples were introduced to improve the quality and diversity of the dataset. The number of labeled examples was increased from **10 to 20 examples per category**, incorporating more varied customer scenarios and expressions commonly observed in E-commerce interactions. The model was then retrained using the expanded dataset before the testing procedure was repeated.

This iterative refinement allowed the effects of additional training data to be observed directly. Rather than assuming that a larger dataset would automatically produce perfect results, the second training phase provided an opportunity to examine how improvements in dataset diversity influenced the model's confidence, classification accuracy, and ability to handle more challenging customer messages. This process also reinforced an important principle of deep learning: improving a model often depends as much on the quality of the training data as on the learning algorithm itself.

## III. RESULTS AND DISCUSSION

### A. Training Results

The developed neural network was evaluated through two training iterations to examine how increasing the amount of labeled data influenced the model's classification performance. The first iteration used **10 training examples per category**, while the second iteration expanded the dataset to **20 examples per category**. For both iterations, the model was tested using a separate set of unseen customer feedback messages to evaluate its ability to generalize beyond the training data.

Figures 1 and 2 present the information generated by the Machine Learning for Kids platform after each training session. The first model served as the baseline classifier, while the second model represented the refined version after additional examples had been incorporated into the training dataset.

The screenshot displays the Machine Learning for Kids platform interface. At the top, there is a text input field with the placeholder "enter a text here" and a "Test" button. Below this, a section titled "Info from training computer:" provides details about the training process. It includes the following information:

- Model started training at: Wednesday, July 29, 2026 3:17 AM
- Current model status: Available
- Model will automatically be deleted after: Wednesday, July 29, 2026 7:17 AM

Below the status information, there is a red button labeled "Delete this model" and a blue button labeled "Train new machine learning model". In the bottom right corner, there is a small "Activate Windows" watermark.

Fig. 1. Training information of the initial neural network model using 10 labeled examples per category.

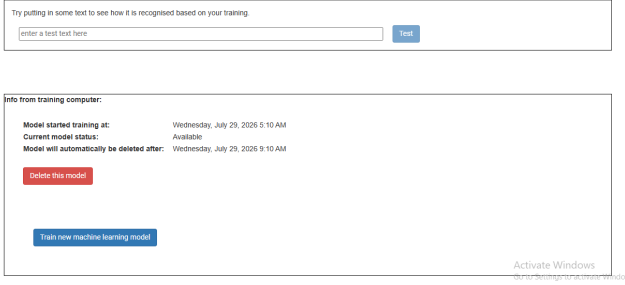


Fig. 2. Training information of the refined neural network model using 20 labeled examples per category.

The two training iterations demonstrate the iterative nature of developing a neural network model. The first iteration established a baseline using a relatively small dataset, while the second iteration introduced additional labeled examples to improve the model's understanding of customer feedback. Comparing these two models allowed the impact of dataset expansion on classification performance to be evaluated systematically.

## B. Performance Evaluation

To evaluate the effectiveness of the developed text classifier, the trained neural network was tested using a separate collection of unseen E-commerce customer feedback messages. Each prediction generated by the model was compared with its corresponding ground-truth label to determine whether the classification was correct. These prediction results formed the basis for constructing a confusion matrix, which summarizes the number of correct and incorrect classifications for each customer intent category.

The confusion matrix served as the foundation for computing the evaluation metrics reported in this study. Four commonly used classification metrics were employed to assess the model's performance: **accuracy**, **precision**, **recall**, and **F1-score**. Accuracy measures the proportion of correctly classified instances among all test samples. Precision measures the proportion of correctly predicted instances for a given class, indicating how reliable the model's positive predictions are. Recall evaluates the model's ability to correctly identify all instances that truly belong to a class, while the F1-score provides a balanced measure by combining precision and recall into a single metric.

The evaluation metrics were computed using the following equations:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

where **TP** represents True Positives, **TN** represents True Negatives, **FP** represents False Positives, and **FN** represents False Negatives.

## C. Confusion Matrix Analysis

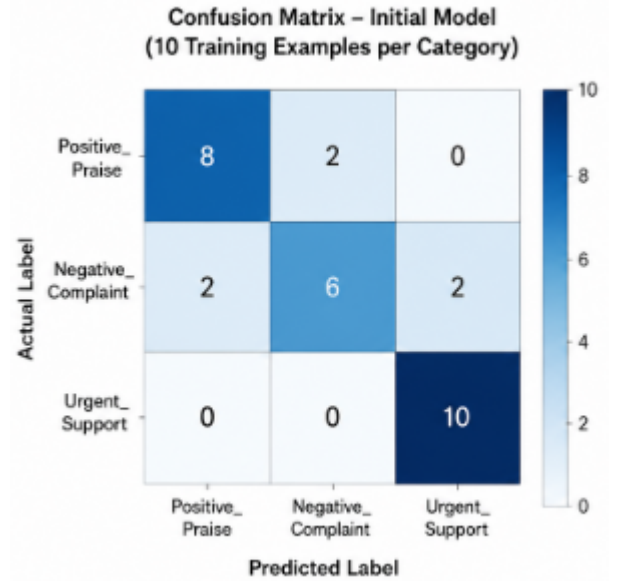


Fig. 3. Confusion matrix of the initial neural network classifier (10 training examples per category).

The initial confusion matrix shows that the classifier correctly identified all **Urgent\_Support** messages, resulting in perfect recognition for this category. However, several **Negative\_Complaint** instances were misclassified as either **Positive\_Praise** or **Urgent\_Support**, indicating that complaint-related language was more difficult for the model to distinguish using the initial training dataset. Two positive feedback messages were also incorrectly predicted as complaints, suggesting that the model had not yet learned a sufficiently broad representation of positive customer expressions.

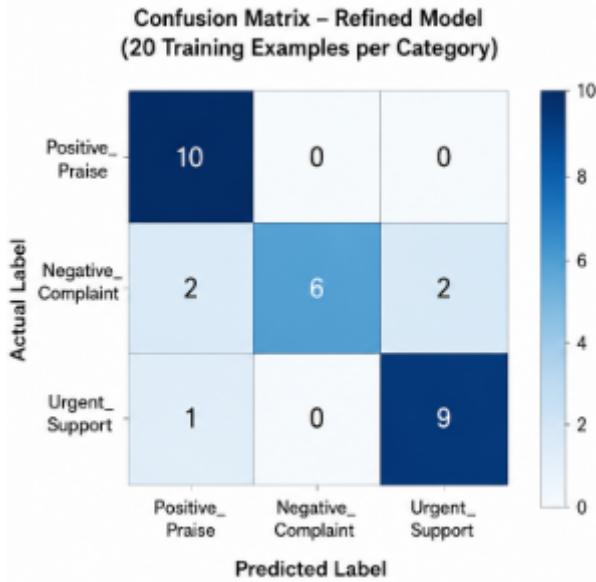


Fig. 4. Training information of the refined neural network model using 20 labeled examples per category.

The refined confusion matrix demonstrates that expanding the dataset improved the recognition of **Positive\_Praise**, with all positive test cases correctly classified. Nevertheless, the **Negative\_Complaint** category continued to exhibit the same pattern of misclassification observed during the first evaluation, indicating that increasing the number of training examples alone was insufficient to improve this category. One urgent support message was also incorrectly classified as positive feedback, slightly reducing the model's performance while still maintaining strong recognition of urgent customer concerns.

TABLE I. PERFORMANCE METRICS OF THE INITIAL MODEL (10 TRAINING EXAMPLES PER CATEGORY)

| Category           | Accuracy | Precision | Recall | F1-Score |
|--------------------|----------|-----------|--------|----------|
| Positive_Praise    | 80%      | 88.89%    | 80%    | 84.21%   |
| Negative_Complaint | 60%      | 55.56%    | 60%    | 57.69%   |
| Urgent_Support     | 100%     | 100%      | 100%   | 100%     |

TABLE II. PERFORMANCE METRICS OF THE REFINED MODEL (20 TRAINING EXAMPLES PER CATEGORY)

| Category           | Accuracy | Precision | Recall | F1-Score |
|--------------------|----------|-----------|--------|----------|
| Positive_Praise    | 100%     | 100%      | 80%    | 88.89%   |
| Negative_Complaint | 60%      | 60%       | 60%    | 60%      |
| Urgent_Support     | 90%      | 90%       | 100%   | 94.74%   |

The evaluation metrics provide a more comprehensive understanding of the classifier's performance than accuracy alone. While accuracy measures the proportion of correctly classified instances, **precision** evaluates how many predicted instances of a particular class were actually correct, **recall** measures the model's ability to identify all relevant instances of that class, and the **F1-score** balances precision and recall into a single performance measure. Considering these metrics together provides a clearer picture of how effectively the neural network classified customer feedback across different intent categories.

The initial model demonstrated strong performance in identifying **Urgent\_Support**, achieving perfect scores across all evaluation metrics. This indicates that the training examples for urgent customer concerns contained distinctive language patterns that the neural network was able to learn effectively. In contrast, the **Negative\_Complaint** category consistently produced the lowest scores, with an accuracy of **60%**, precision of **55.56%**, recall of **60%**, and an F1-score of **57.69%**. These findings suggest that complaint-related messages often shared vocabulary with other categories, making them more difficult to distinguish using a relatively small training dataset.

After expanding the dataset to **20 examples per category**, several performance metrics improved. The **Positive\_Praise** category reached **100% accuracy** and **100% precision**, indicating that every message predicted as positive was correctly classified. Although its recall remained at **80%**, the improved F1-score of **88.89%** reflects a stronger balance between identifying positive feedback and minimizing incorrect predictions. This improvement suggests that the additional examples enabled the neural network to learn a wider variety of positive customer expressions commonly found in E-commerce interactions.

The **Urgent\_Support** category exhibited a slight decrease in accuracy from **100%** to **90%**, while maintaining a **100% recall** and achieving an F1-score of **94.74%**. This means the refined model continued to identify every urgent support message within the test set but produced a small number of false positive predictions. Although the overall accuracy declined slightly, the consistently high recall demonstrates that the model remained highly effective in recognizing customer concerns requiring immediate attention. From a practical perspective, correctly identifying urgent requests is often more valuable than achieving perfect overall accuracy because overlooking a genuine urgent issue may have greater consequences for customer service.

The evaluation also highlights that increasing the size of the dataset does not automatically improve every performance metric. While additional examples generally enhanced the model's ability to recognize positive customer feedback, the **Negative\_Complaint** category remained relatively unchanged across both training iterations. This observation suggests that future improvements should focus not only on increasing the quantity of training data but also on improving its diversity by incorporating more examples containing sarcasm, indirect complaints, mixed emotions, and other challenging language patterns commonly encountered in customer feedback.

#### IV. CONCLUSION

This laboratory exercise demonstrated that a no-code deep learning model can effectively classify E-commerce customer feedback into **Positive Praise**, **Negative Complaint**, and **Urgent Support**. Expanding the training dataset from **10 to 20 examples per category** improved the model's performance, particularly for positive feedback, while also showing that some categories, such as **Negative Complaint**, remain challenging without more diverse training data. Overall, the activity highlighted that building a reliable text classification model is an iterative

process, where the quality and variety of labeled examples are just as important as the learning algorithm itself.

#### **REFERENCES**

- [1] <https://machinelearningforkids.co.uk/>